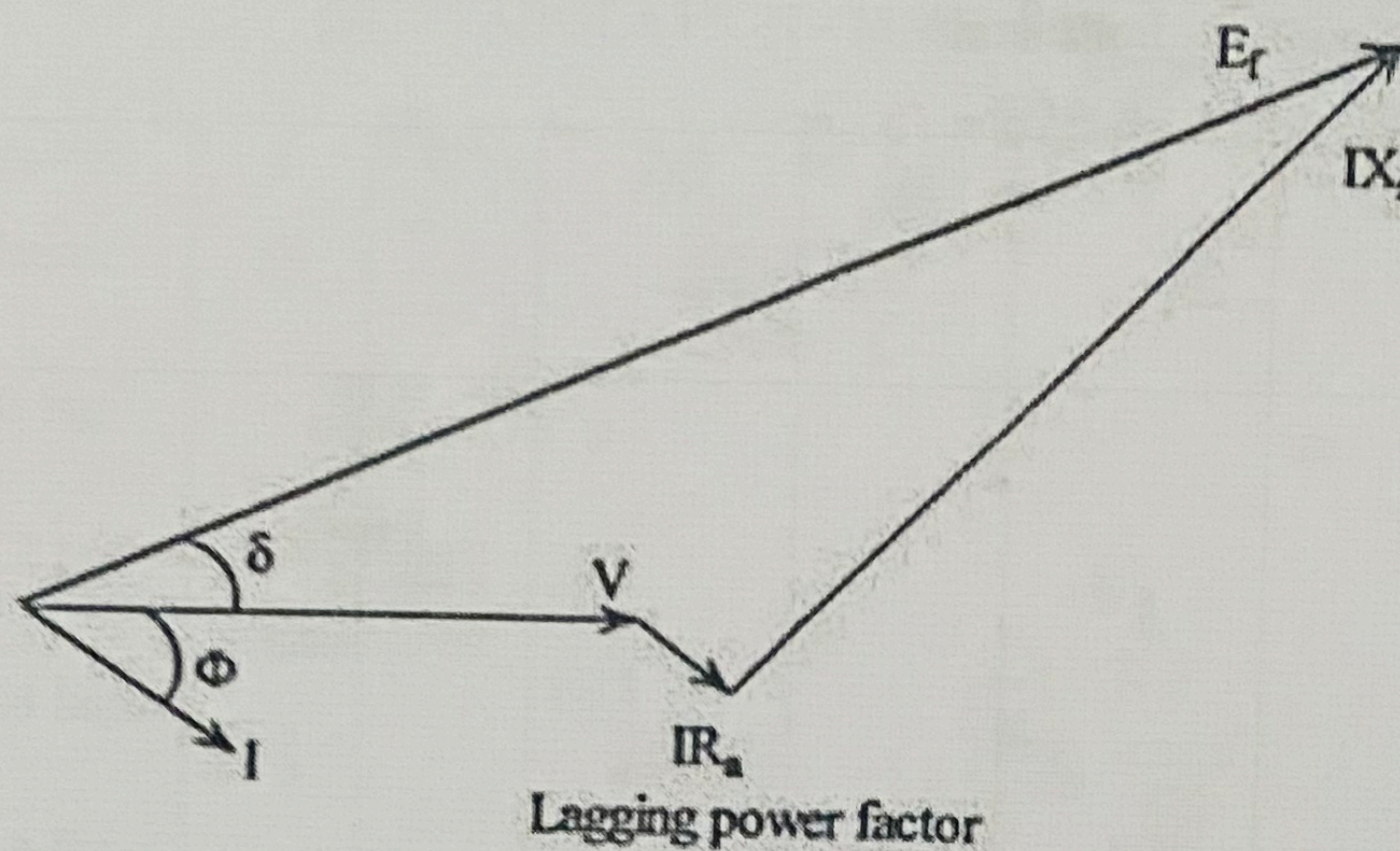




SAMPLE CALCULATION

MMF METHOD

Regulation at full load and ____ pf. Lag

$$\bar{V} = V \angle 0^\circ = \underline{\hspace{2cm}} V$$

$$\cos \phi = \underline{\hspace{1cm}} \text{ lag}$$

$$\bar{I} = I \angle (-\phi)^\circ = \underline{\hspace{2cm}} A$$

$$\bar{E}' = V \angle 0^\circ + I \angle -\phi^\circ \times R_a = E' \angle (-\sigma)^\circ = \underline{\hspace{2cm}} V$$

Refer OCC and find I_{fr} corresponding to E' .

$$\bar{I}_{fr} = I_{fr} \angle (-\sigma + 90^\circ) = \underline{\hspace{2cm}} A$$

I_{fa} is the field current required to circulate rated current on short circuit (from SC test)

$$\bar{I}_{fa} = I_{fa} \angle (-\phi + 180^\circ) \text{ for lag } (\bar{I}_{fa} = I_{fa} \angle (+\phi + 180^\circ) \text{ for lead})$$

$$\bar{I}_f = \bar{I}_{fr} + \bar{I}_{fa} = \underline{\hspace{2cm}} A = I_f \angle (90 + \delta)^\circ$$

Hence, $I_f = \underline{\hspace{2cm}} A$, $\delta = \underline{\hspace{1cm}}^\circ$

Refer OCC and find E_f corresponding to I_f

$$\bar{E}_f = E_f \angle \delta^\circ = \underline{\hspace{2cm}} V$$

$$\% \text{ regulation} = \frac{E_f - V}{V} \times 100 \% = \underline{\hspace{2cm}} \%$$